

The Pathology Laboratory

Introduction · Normal Organs · Atrophy, Hypertrophy and Metaplasia

PART 1 How to Work in the Pathology Lab

PART 2 The Normal Baseline — Heart, Lung, Kidney

PART 3 Cellular Adaptation — the Concepts

PART 4 This Session's Specimens

PART 5 Homework, Self-Test and Glossary

SYNTHESIZED FROM

“Pathological Experiment-1: Introduction”

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**Robbins & Kumar *Basic Pathology*, 11e (Ch. 1–2) · Robbins & Cotran *Pathologic Basis of Disease*
· Wheater's *Functional Histology***

For the mechanisms the slides state but do not explain

How to Use This Guide

A pathology lab is not a lecture with pictures. It has one skill at its centre: **look at a thing and say what you see, in an order, in words another doctor would accept**. The diagnosis is the easy part — it falls out of the description. This guide is built around that skill.

The organising idea

Every lesion in this session is a **cell changing size or identity because its circumstances changed**. Less work, less blood or less hormone → the cell shrinks (atrophy). More work → it enlarges (hypertrophy). A hostile surface → it changes into a cell type built to survive there (metaplasia). Learn the three stimuli and you do not have to memorise the appearances; you can predict them.

The two tags

Every observation in this guide is marked for where it comes from, because the two are examined separately and described differently:

- **GROSS** — what you can see with the naked eye on the specimen: size, shape, colour, surface, cut surface.
- **MICRO** — what appears down the microscope: architecture at low power, cells at high power.

The five kinds of box

Box	What it contains
High-Yield	Repeatedly examined. If you memorise nothing else, memorise these.
Exam Trap	Where two concepts are confused, or the intuitive answer is wrong.
Bench Technique	How to actually do it: which objective, which order, what to write down.
Clinical Correlation	What the patient with this specimen looked like in life.
Description Template	A numbered order to follow at the bench, with the question to ask at each step.

The specimen card

Part 4 gives every specimen on this session's list the same card: **what you see → why it looks like that → diagnosis → the sentence to write in your report**. Cover the lower half of the card and try to produce it yourself; that is the exam task, exactly.

Suggested study sequence

1. **Before the lab (≈30 min)**. Read Part 1 and Part 3. Go in knowing the method and the four words.
2. **At the bench**. Keep Part 2 open beside the microscope — you cannot call something abnormal until you know what normal looks like.
3. **Writing the report (≈45 min)**. Work through Part 4, one card per specimen, and copy the structure of the model reports in §1.4.
4. **Night before the exam (≈20 min)**. Part 5 — the self-test, then the glossary.

A NOTE ON THE SOURCES

The slide deck is a *list* — specimen names, a few descriptive sentences, and photographs. It states what to look at but rarely why the tissue looks that way. Everything on the list is here in full, in the lecturer's own words where he gave them; the mechanisms underneath are filled in from Robbins. Where the deck's own wording is ambiguous — most notably the phrase “central hypertrophy” on the cardiac specimen — the guide says so and gives you both readings.

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PART ONE

01

How to Work in the Pathology Lab

This Part is the method, and it is worth more marks than any single specimen. A pathologist is paid to convert a *thing* into *words* — reliably, in the same order every time, so that a colleague reading the words can reconstruct the thing. Learn the order once and it serves you for every specimen you will ever see.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Describe any gross specimen in a fixed three-step order, without omissions.
- Use the microscope in the correct sequence and say what each magnification is for.
- Write a lab report in the format the department marks.
- Distinguish a *description* from a *diagnosis* — and know why writing the second without the first loses marks.

1.1 What This Course Is, and How It Is Marked

The laboratory course runs from **16 September to 11 November** and is taught by **Dr Yongtao He** (Attending, Pathology Department, Sir Run Run Shaw Hospital), whose clinical work is *surgical pathological diagnosis* — which is to say, the person marking your descriptions does this for a living.

Five methods make up the course. They are listed in the order of increasing difficulty, and this session covers the first two:

TABLE 1.1 THE FIVE METHODS OF THE LABORATORY COURSE

Method	What it asks of you
Observe and describe gross specimens	Naked-eye examination of the organ. §1.2
Observe and describe microscopic slides	Tissue architecture and cells. §1.3
Pathological and differential diagnosis	Name the lesion — and name what it is <i>not</i> , and why.
Clinico-pathological conference (CPC)	Reconcile the clinical story with the tissue. Later in the course.
Autopsy	The whole body as one specimen. Later in the course.

BENCH TECHNIQUE — WHAT A REPORT MUST CONTAIN

Every experiment report (the homework) has three fixed parts:

- **1–2 slides** that you must **draw**.
- **1–2 specimens** that you must **describe**.
- Submission on **Xuezaizheda** (学在浙大) → **Homework**, as a Word document with your photographs taken from the **Digital Human STEM Morphology System**.

EXAM TRAP — DESCRIPTION IS NOT DIAGNOSIS

The commonest way to lose marks is to write “*Brown atrophy of the heart*” and stop. That is the **conclusion**. The marks are in the **evidence**: reduced size, narrowed transverse diameter, light brown colour, tortuous coronaries. A correct diagnosis with no description is an unsupported guess; a full description with a wrong diagnosis still earns most of the marks. **Always describe first, then name.**

1.2 Describing a Gross Specimen

Three steps, always in this order. The order matters because it moves from **certain to uncertain**: you can be sure what organ you are holding long before you can be sure what is wrong with it, and a description that starts at the lesion has no frame of reference.

DESCRIPTION TEMPLATE — GROSS SPECIMEN

1 Identify the organ.

ASK What is this? If unsure, say what it resembles and what structures you can identify — an unnamed organ still earns description marks.

2 Describe the whole organ — size and weight · shape · colour · surface · cut surface.

ASK Is it bigger or smaller than normal? Is the shape preserved? Is the capsule smooth or granular? What does the cut face show?

3 Describe the lesion — localisation · number · size · relationship to the surrounding normal tissue.

ASK Where exactly? One or many? How big? Is the border sharp or blended — and does it push the normal tissue aside or replace it?

Step 2 and step 3 are often confused. The rule is simple: **step 2 is about the organ, step 3 is about what has gone wrong inside it.** In a diffuse process such as atrophy the two merge, because the whole organ is the lesion — and in that case say so explicitly rather than leaving step 3 blank.

HIGH-YIELD — THE SIX WORDS FOR ANY ORGAN

Size · Weight · Shape · Colour · Surface · Cut surface. Run down that list for every gross specimen and you physically cannot omit anything the marker is looking for. If a feature is normal, say that it is normal — “*the capsule is smooth and the surface is not nodular*” is a positive observation, not a gap.

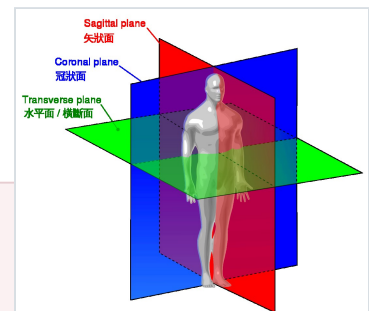


Fig 1.1 The three planes. A specimen is almost always cut in one of them, and saying which is part of describing the cut surface — a kidney bisected in the **coronal** plane shows cortex, medulla and pelvis at once, which is why that is the standard renal cut.

1.3 Reading a Microscopic Slide

The microscope has its own fixed order, and it exists to stop you doing the thing every beginner does — starting at 40× and getting lost in beautiful cells with no idea which part of which organ you are looking at.

DESCRIPTION TEMPLATE — MICROSCOPIC SLIDE

1 Naked eye, slide held to the light.

ASK What shape is the tissue on the glass? Where are the pale and dark zones? This tells you the organ and where to point the objective — a kidney shows cortex and medulla, a stomach shows a mucosal ribbon.

2 Low power, 4× or 10× — orientation and identification.

ASK What organ, and which layers, in what order? Is the architecture preserved or destroyed? **Every diagnosis is made here.** Then scan the whole section before you go higher.

3 High power, 40× — fine detail only.

ASK Now, and only now: what do individual cells look like? Nucleus, cytoplasm, pigment, inclusions. High power *confirms* a diagnosis; it does not make one.

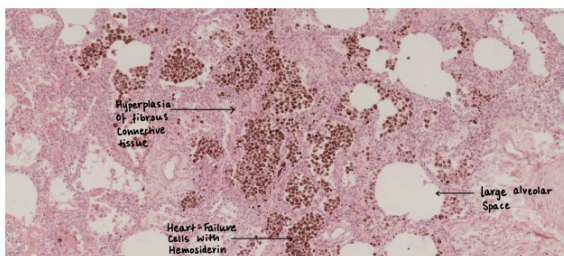
EXAM TRAP — “I CAN’T FIND THE LESION”

Almost always this means you went to 40× too early. At 40× the field is a few hundred micrometres across; a lesion that occupies a quarter of the section is then invisible, because you are inside it and everything looks uniform. Go back to 4×. If the whole section looks abnormal at low power, that is the finding — it means the process is **diffuse**, which is itself diagnostic information.

1.4 Writing the Report — Two Model Answers

The department gave two worked examples. Copy their **shape** exactly: a heading naming the specimen, then **Organ or Tissue**, then **Description (Evidence of Diagnosis)** as bullet points, then **Pathological Diagnosis** on its own line. The phrase in brackets is the whole philosophy — your bullets are not decoration, they are the evidence that entitles you to the diagnosis.

1. Microscopic Slide 1



Organ or Tissue: Lung

Description (Evidence of Diagnosis):

- Alveoli wall thickened due to proliferation and accumulation of connective tissue (fibroblasts)
- Visible large alveolar spaces
- Evidence of hyperplasia of fibrous connective tissue in the alveolar wall
- Presence of heart failure cells, with yellow brown granules (hemosiderin), in the cytoplasm
- No evidence of dilation in the capillaries

Pathological Diagnosis: Chronic Congestion of the Lungs

Fig 1.2 Model report for a **microscopic slide**. Note that one bullet is a negative finding — “no evidence of dilation in the capillaries”. Recording what is absent is what separates a description from a list.

3. Gross-Specimen 26



Organ or Tissue: Liver

Description (Evidence of Diagnosis):

- Liver appears slightly enlarged and firm
- Visible dark lines due to hemorrhaging near engorged sinusoids
- Those dark lines indicate congestion
- Dark and white (unaffected) areas alternate causing a nutmeg appearance
- Yellow-grey line indicates a fatty change of the platelets

Pathological Diagnosis: Chronic Congestion of the Liver (Nutmeg Liver)

Fig 1.3 Model report for a **gross specimen**. The bullets move outward-in — whole organ first (“slightly enlarged and firm”), then the pattern on the cut surface, then the interpretation.

HIGH-YIELD — THE FOUR HEADINGS, MEMORISED

1. Specimen number and type · **2.** Organ or Tissue · **3.** Description (Evidence of Diagnosis) · **4.** Pathological Diagnosis. Four to six bullets under heading 3 is the right length — enough to support the diagnosis, short enough that every bullet earns its place.

BENCH TECHNIQUE — WRITE IN OBSERVABLE LANGUAGE

A description should contain nothing that requires a diagnosis to have been made. Write “*yellow-brown granular pigment in the perinuclear cytoplasm*”, not “*lipofuscin*”; write “*the renal pelvis and calyces are markedly dilated and the parenchyma is thinned*”, not “*hydronephrosis*”. Save the names for the diagnosis line. This is not pedantry — it is the difference between evidence and assertion, and markers look for it specifically.

PART TWO

02

The Normal Baseline

You cannot call a heart small until you know what a heart weighs. Half of pathology is anatomy and histology held firmly enough that a deviation announces itself. This Part is the reference you keep open beside the microscope — the normal heart, lung and kidney, gross and microscopic, with the numbers the lecturer gave.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- State the normal weight, size, shape and colour of the heart, lung and kidney.
- Name the three layers of the heart wall and identify each down the microscope.
- Recognise an alveolus, an interalveolar septum and an alveolar capillary.
- Identify glomerulus, tubule and interstitium in renal cortex, and say which is which.

2.1 Heart

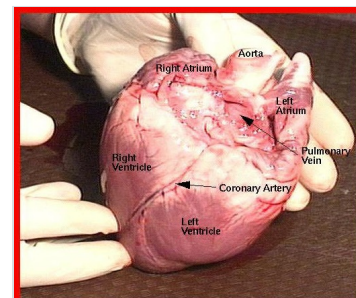


Fig 2.1 The normal heart in the fresh state, anterior view. Identify the four chambers and the coronary artery running in the interventricular groove — a vessel that will matter in §4.1, where it becomes tortuous.

TABLE 2.1 NORMAL HEART — THE SIX WORDS

Feature	Normal
Shape	Peach-like — a blunt cone with the apex pointing left, down and forward
Colour	Yellow and red — yellow from the epicardial fat, red from the muscle
Size	About the size of the subject's own fist
Weight	Male $\approx$ 280 g · Female $\approx$ 240 g
Site	Left thoracic cavity, in the middle mediastinum
Function	A pump — supplies blood to organs, tissues and cells

The three layers, outside in

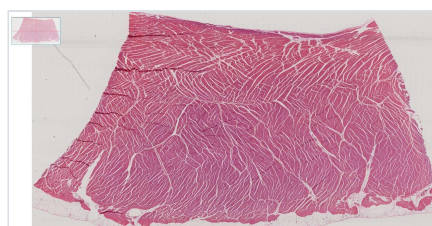


Fig 2.2 The heart wall at 4 $\times$. Before anything else, find the three layers and their order. Everything else on this slide is detail.

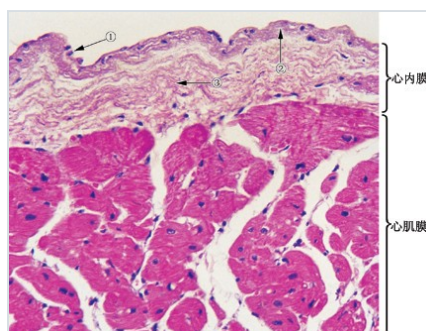


Fig 2.3 Endocardium (心内膜, upper) over myocardium (心肌膜, lower). The endocardium is thin, pale and has a flat endothelial lining facing the chamber.

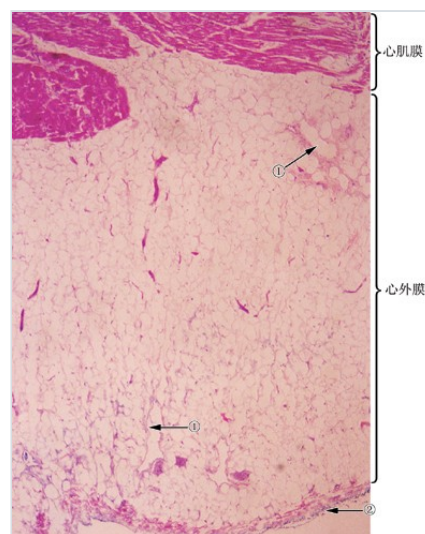


Fig 2.4 Myocardium (upper) passing into epicardium (心外膜, lower) — recognisable immediately by its adipose tissue and vessels.

TABLE 2.2 HEART WALL — WHAT TO LOOK FOR IN EACH LAYER

Layer	Contains	How you recognise it
Endocardium — internal	Endothelial layer ; a subendothelial layer of smooth muscle fibres; branches of the impulse-conducting system (Purkinje fibres)	Thin, pale band lining the chamber. Purkinje fibres are large, pale, glycogen-rich cells sitting just beneath it — bigger and emptier-looking than working myocytes
Myocardium — middle	Striated (striped) muscle , branching, with central nuclei and intercalated discs	The bulk of the wall. Fibres branch and rejoin — the feature that distinguishes cardiac from skeletal muscle
Epicardium — external	Mesothelium (simple squamous epithelium) over a thin layer of connective tissue containing arterioles, venules, capillaries, nerves and adipose tissue	Fat is the giveaway — a pale, bubbly layer on the outer surface. The mesothelium itself is a single flat cell layer you may need 40× to see

EXAM TRAP — WHICH SURFACE IS WHICH

On a slide with no chamber visible, students guess. Use the two reliable cues instead: **fat means epicardium** (the outside), and **a flat lining cell layer with no fat means endocardium** (the inside). If you see both in one field, the section runs the full thickness of the wall — say so, because it tells the reader the orientation of the block.

2.2 Lung

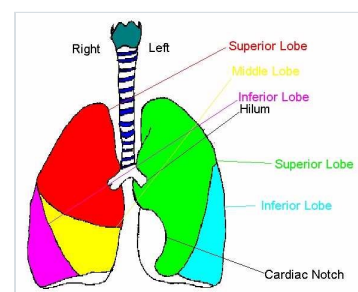


Fig 2.5 Lobes and the cardiac notch. **Right: three lobes. Left: two**, because the heart occupies the space where a middle lobe would be. If a specimen has three lobes it is a right lung — that is a free identification mark.

TABLE 2.3 NORMAL LUNG — THE SIX WORDS

Feature	Normal
Shape	Durian-like — a half-cone, flat on the mediastinal face, domed over the diaphragm
Colour	Grey and red ; grey-black stippling with age from inhaled carbon (anthracosis) is normal in a city dweller
Size and weight	Various — the lecturer's own word. Lung weight depends heavily on how much blood and oedema fluid it holds
Site	Thoracic cavity, one either side of the mediastinum
Function	Gas exchange — oxygen in, carbon dioxide out
Texture	Spongy and crepitant — it crackles under the fingers because it contains air. Loss of crepitus is a real finding

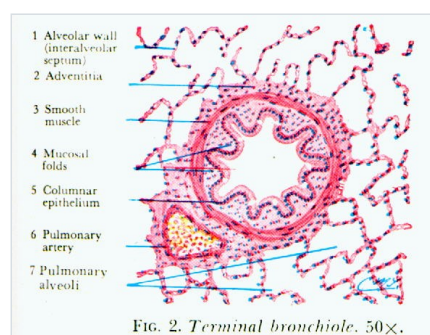


Fig 2.6 Terminal bronchiole, 50x. Still a conducting tube: a complete columnar epithelium, mucosal folds and a ring of smooth muscle. No alveoli open off its wall.

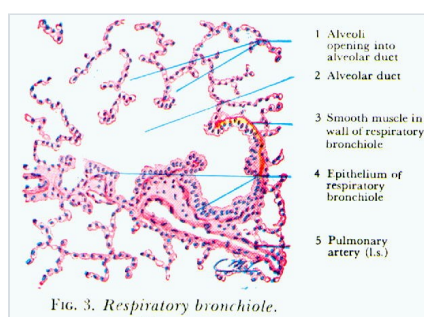


Fig 2.7 Respiratory bronchiole. The wall is now interrupted by alveoli opening directly off it — the transition from conducting to respiratory airway, and the definition of the acinus.

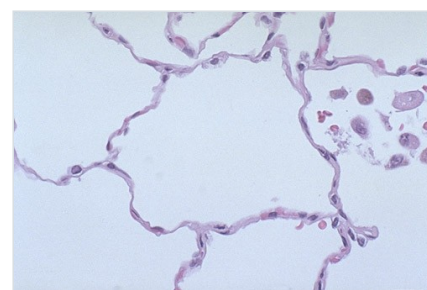


Fig 2.8 Normal alveoli. The spaces are empty, the **interalveolar septa** are hair-thin, and the **capillaries** within them are barely visible. Any thickening of these septa is abnormal.

HIGH-YIELD — THREE STRUCTURES, AND THAT IS ALL

For this session the lecturer labels exactly three things in the lung: the **alveolus** (the air space), the **interalveolar septum** (the wall between two of them) and the **capillary** (inside that wall). Almost every lung pathology you will meet is a change in one of the three — fluid or cells *in* the alveolus, thickening of the septum, congestion of the capillary. Fix the normal appearance of all three now and the rest of the course becomes comparison.

2.3 Kidney

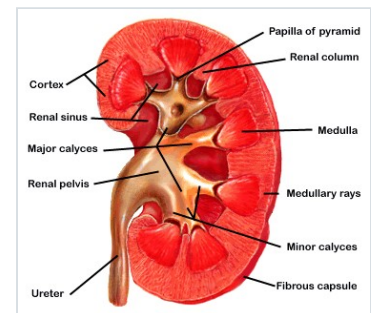


Fig 2.9 Coronal section. Learn this order outward-in: **fibrous capsule** → **cortex** → **medulla (pyramids)** → **papilla** → **minor calyx** → **major calyx** → **renal pelvis** → **ureter**. In §4.1 an obstruction below the pelvis will dilate everything above it, in exactly this sequence.

TABLE 2.4 NORMAL KIDNEY — THE SIX WORDS

Feature	Normal
Shape	Like a broad bean — convex laterally, concave at the hilum
Colour	Red — it receives about a fifth of the cardiac output
Size	10–12 × 5–6 × 3–4 cm
Weight	125–150 g each
Site	Abdominal cavity — retroperitoneal, on the posterior wall
Function	Urine formation ; elimination of toxins

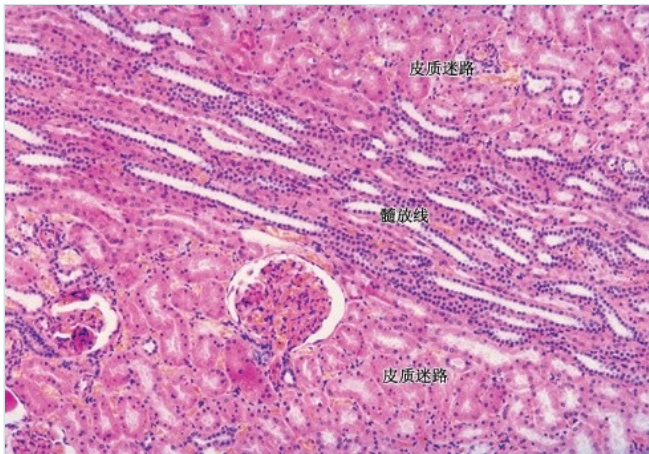


Fig 2.10 Renal **cortex** at low power. The cortical labyrinth (皮质迷路) carries the glomeruli; the **medullary ray** (髓放线) is the straight bundle of tubules running down towards the medulla — a useful landmark for orientation.

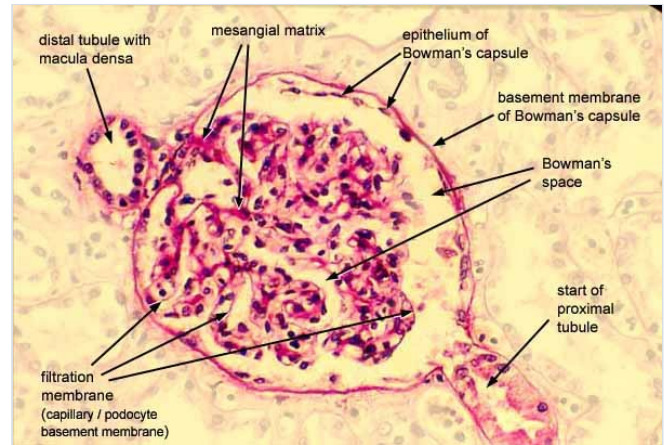


Fig 2.11 A normal **glomerulus**: the capillary tuft, **Bowman's space** around it, the epithelium of Bowman's capsule, mesangial matrix, and the macula densa of the distal tubule alongside.

For describing purposes the renal cortex has exactly three components, and every renal slide question is really asking which of them has changed:

- **Glomerulus** — a capillary network inside Bowman's capsule. The filter.
- **Renal tubules** — the great majority of the tissue by volume. Proximal tubules have a brush border and deep pink cytoplasm; distal tubules are paler with a clearer lumen.
- **Interstitialium** — the sparse connective tissue and capillaries between tubules. Normally you can barely see it; when you *can* see it, something is wrong.

2.4 Normal Numbers Worth Knowing

TABLE 2.5 THE REFERENCE FIGURES FROM THIS SESSION

Organ	Normal	Where it is used in this guide
Heart weight	M 280 g · F 240 g	Both directions: below it in atrophy (§4.1), above it in hypertrophy (§4.2)
Heart size	The subject's own fist	The bedside estimate, since a specimen rarely arrives with its owner's hand
Kidney weight	125–150 g	Compensatory hypertrophy can take one kidney well above this
Kidney size	10–12 × 5–6 × 3–4 cm	Note the trap in §4.1: an obstructed kidney is <i>enlarged</i> while its parenchyma is <i>atrophied</i>
Lung lobes	Right 3 · Left 2	Side identification on any pulmonary specimen
Heart wall layers	Endo- → myo- → epicardium	Inside to outside. Asked directly, and needed to orient any cardiac slide

BENCH TECHNIQUE — THE OTHER FOUR ORGANS

The specimen list for this session also includes normal **liver, spleen, stomach and intestine**, both as gross specimens and as slides, but the deck does not describe them. Apply the same six words and the same three microscope steps. Two orientation cues will get you started: the **spleen** shows white pulp nodules scattered in red pulp at low power, and the **stomach and intestine** are recognised by their layered wall — mucosa, submucosa, muscularis propria, serosa — with the mucosal pattern (gastric pits versus intestinal villi) telling you which.

PART THREE

03

Cellular Adaptation — the Concepts

Four words carry this session: **atrophy, hypertrophy, hyperplasia, metaplasia**. They are not four unrelated definitions to memorise. They are four answers to one question — *what does a cell do when its demands change?* — and once you can state the question a cell is answering, you can predict which answer it will give.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Define atrophy, hypertrophy, hyperplasia and metaplasia, and give the stimulus for each.
- Explain why a cell shrinks rather than dies, and name the two machines that remove its bulk.
- Say why the heart can only hypertrophy and the liver can also hyperplase.
- Explain why metaplasia is protective and dangerous at the same time.

3.1 Adaptation, Injury and Death — One Spectrum

A cell in its normal state sits in a narrow band of demand. When demand moves outside that band the cell has three possible fates, and which one it gets depends entirely on **how severe the change is and how fast it arrives**:

1

Adaptation — a new, stable steady state. The cell survives, changed. This session is entirely about adaptation.

2

Reversible injury — the cell is overwhelmed but not yet past the point of no return. Swelling and fatty change. Next week's session.

3

Irreversible injury and death — necrosis or apoptosis.

HIGH-YIELD — THE SINGLE SENTENCE

Adaptation is a cell successfully avoiding death by changing what it is. That is why an atrophic cell is not a dying cell and why the distinction is examined: the atrophic cell has *reduced its requirements to match its supply*, which is a solution, not a failure.

TABLE 3.1 THE FOUR ADAPTATIONS — STIMULUS, CHANGE, EXAMPLE

Adaptation	What changes	Stimulus	Example in this session
Atrophy	Cell size ↓ (and often number ↓) → organ smaller	Less work, less blood, less nutrition, less hormone, less nerve supply, ageing, pressure	Brown atrophy of heart; brain atrophy; obstructed kidney; leprotic limb
Hypertrophy	Cell size ↑, number unchanged → organ larger	Increased workload, or trophic hormonal signal	Cardiac hypertrophy; compensatory renal hypertrophy
Hyperplasia	Cell number ↑ → organ larger	Hormonal drive or compensatory demand — only in tissues that can divide	Part of the renal compensatory response; the fibroblasts in the model lung report
Metaplasia	One mature cell type replaced by another	Chronic irritation or a persistently hostile local environment	Gastric mucosa with intestinal metaplasia

3.2 Atrophy

DEFINITION

Atrophy is a **decrease in the size of a cell, tissue, organ or multiple organs**, associated with a variety of pathological conditions such as abnormal cellular changes, **ischaemia, malnutrition or hormonal changes**.

Note what the definition does *not* say. It does not say cells die. Atrophy is primarily a reduction in the **size** of each surviving cell; cell number may also fall, by apoptosis, but the shrunken cell is alive and functioning at a lower level.

○ MECHANISM — HOW A CELL MAKES ITSELF SMALLER

- 1 The stimulus arrives: **reduced workload, blood supply, innervation, nutrition or hormonal stimulation**.
WHY: All five converge on one message — the cell is being asked to do less, or given less to do it with.
- 2 Protein **synthesis falls** and protein **degradation rises**.
WHY: The cell cannot survive long in negative balance unless it shrinks to match. Reducing bulk is how it lowers its own running costs.
- 3 The **ubiquitin–proteasome pathway** tags unneeded cytosolic proteins and destroys them.
WHY: This is the route that removes the contractile proteins of a disused muscle — which is why a limb in a cast loses bulk within weeks.
- 4 **Autophagy** increases: the cell encloses its own organelles in vacuoles and digests them with lysosomal enzymes.
WHY: This is a starving cell eating its own furniture for fuel. It is a survival strategy, and it is the source of the next step.
- 5 Some digested membrane lipid **cannot be broken down completely**, and the residue accumulates as **lipofuscin** granules.
WHY: Lipofuscin is literally the ash left over from autophagy. This single fact explains the colour of the organ in §4.1 — a heart full of ash is a brown heart.
- 6 The shrunken cell reaches a **new, smaller steady state** and survives there indefinitely.
WHY: If the stimulus is removed early the cell regrows. If it persists, cells eventually die by apoptosis and the organ shrinks further — which is why chronic atrophy is only partly reversible.

NET RESULT A smaller cell with fewer organelles, a lower metabolic rate, brown lipofuscin residue — and, for as long as the stimulus lasts, a smaller organ that still works.

TABLE 3.2 CAUSES OF ATROPHY, WITH THIS SESSION'S EXAMPLES

Cause	Typical setting	Specimen here
Decreased workload (dis-use)	Limb immobilised in plaster; prolonged bed rest	Contributes to the leprotic limb
Loss of innervation	Peripheral nerve injury; poliomyelitis; leprosy	Specimen 4 — leprotic atrophy of the lower limb
Diminished blood supply (ischaemia)	Gradual arterial narrowing, most often atherosclerosis	Specimen 2 — brain atrophy
Inadequate nutrition	Marasmus; cancer cachexia; chronic illness	Specimen 2 is attributed to lack of nutrients
Loss of endocrine stimulation	Endometrium and breast after the menopause	—
Pressure	An expanding mass or a fluid collection compressing tissue	Specimen 3 — renal compression atrophy

Cause	Typical setting	Specimen here
Ageing (senile atrophy)	Brain and heart, in everybody, slowly	Specimen 1 — brown atrophy of the heart

EXAM TRAP — ATROPHY VERSUS HYPOPLASIA VERSUS APLASIA

All three give a small organ, and they are told apart by **when** the problem occurred, not by how the organ looks. **Atrophy** — the organ grew to normal size and then shrank. **Hypoplasia** — it never reached normal size (a developmental failure). **Aplasia** — it never formed at all. This distinction decides §4.2: in the compensatory renal specimen the *small* kidney is **hypoplastic**, not atrophic, because it was congenitally dysplastic and never developed properly.

3.3 Hypertrophy

DEFINITION

Hypertrophy is an **increase in the size of cells**, with **no increase in their number**, producing an increase in the size of the organ. It occurs in tissues that cannot divide, and in dividing tissues it accompanies hyperplasia.

○ MECHANISM — PRESSURE OVERLOAD TO A THICKER WALL

- A sustained **increase in workload** — for the left ventricle, most often systemic hypertension or aortic stenosis.

WHY: *The demand must be sustained. A single sprint does not remodel a heart; years of pumping against a raised afterload do.*
- Mechanical stretch and stress on the myocyte activate **mechanosensors**, and locally released growth factors and vasoactive agents amplify the signal.

WHY: *The cell has no way to measure blood pressure directly; it measures the tension in its own membrane and cytoskeleton, and treats that as the input.*
- Signal transduction reaches the nucleus and switches on a **gene programme**: more contractile proteins, and re-expression of **foetal genes** such as atrial natriuretic factor and β -myosin heavy chain.

WHY: *The adult heart reverts to an embryonic gene set because those isoforms contract more economically — more force per unit of ATP, at the cost of speed.*
- Each myocyte accumulates more **sarcomeres and mitochondria**. It gets bigger; it does not divide.

WHY: *Cardiac muscle is a **permanent tissue** — its cells left the cell cycle before birth. Hypertrophy is the only growth option available to it, which is why the heart can never hyperplase.*
- The wall thickens and, for a period, the heart **compensates** — normal output against an abnormal load.

WHY: *This is the useful phase, and it can last decades.*
- Eventually the enlarged myocyte **outgrows its blood supply**: capillary density does not keep pace with muscle mass.

WHY: *The coronary circulation was built for the original muscle. A cell twice the size sits twice as far from its nearest capillary, so the subendocardium — the furthest territory — becomes chronically ischaemic.*

NET RESULT Compensated hypertrophy first, then myocyte degeneration, interstitial fibrosis, chamber dilatation and **heart failure**. Hypertrophy is an adaptation with an expiry date.

HIGH-YIELD — CONCENTRIC VERSUS ECCENTRIC

Concentric hypertrophy — **pressure** overload (hypertension, aortic stenosis). Sarcomeres are added *in parallel*, the wall thickens and the chamber cavity is **normal or small**.

Eccentric hypertrophy — **volume** overload (aortic or mitral regurgitation). Sarcomeres are added *in series*, the wall thickens modestly and the chamber is **dilated**.

The rule in one line: **pressure thickens, volume dilates**.

3.4 Hyperplasia — and Why It Is Not Hypertrophy

DEFINITION

Hyperplasia is an **increase in the number of cells** in an organ, resulting from proliferation. It can only occur in tissues whose cells are **capable of dividing**.

Hypertrophy and hyperplasia both enlarge an organ, and they frequently happen together — the pregnant uterus does both. Which one a tissue can use is decided not by the stimulus but by the **proliferative capacity of the tissue itself**:

TABLE 3.3 TISSUE TYPE DECIDES WHICH ADAPTATION IS AVAILABLE

Tissue type	Examples	Can it hypertrophy? Hyperplase?
Labile	Surface epithelia, bone marrow, gut mucosa	Both — but hyperplasia dominates; these cells divide continuously
Stable	Liver, kidney tubule, fibroblast, smooth muscle	Both — normally quiescent, but re-enter the cycle when called on
Permanent	Cardiac muscle , skeletal muscle, neurons	Hypertrophy only . The cells left the cell cycle and cannot return

EXAM TRAP — THE ENLARGED KIDNEY

“Compensatory hypertrophy of the kidney” is the name on the specimen, and it is the name you should write. But the kidney’s response to losing its partner is in fact **both** hypertrophy of existing tubular cells *and* hyperplasia of some of them — the tubular epithelium is a **stable** tissue and retains the ability to divide. The heart, being permanent, does not have that option. If a question asks which organ can only hypertrophy, the answer is the heart, never the kidney.

3.5 Metaplasia

DEFINITION

Metaplasia is a **reversible change in which one differentiated cell type is replaced by another** differentiated cell type, better able to withstand the local environment.

MECHANISM — WHY THE TISSUE CHANGES ITS MIND

- 1 A **chronic irritant** acts on the surface — gastric acid and bile in the stomach, cigarette smoke in the bronchus, acid reflux in the oesophagus.
WHY: The injury has to be chronic. An acute insult causes inflammation or necrosis, not metaplasia; the tissue needs time to re-plan.
- 2 The existing mature cells are poorly suited to survive it and are steadily lost.
WHY: Gastric surface epithelium is built to secrete, not to resist prolonged chemical assault.
- 3 **The differentiated cells do not transform.** Instead the **tissue stem cells** are **reprogrammed**, by signals from the damaged environment, to follow a different differentiation pathway.
WHY: This is the single most important point in the topic and the most commonly misunderstood. Metaplasia is a change in what the stem cell makes, not a mature cell changing into something else.
- 4 The new epithelium is **tougher** and the tissue survives.
WHY: Intestinal epithelium with goblet cells and an absorptive brush border tolerates a chemically hostile lumen far better than gastric surface epithelium does. The adaptation works.
- 5 But the new epithelium is in the **wrong place**, and the specialised function of the original is lost.
WHY: Metaplastic bronchial squamous epithelium is durable but has no cilia and no mucus — so the airway loses its escalator exactly where it most needs one.
- 6 If the irritant persists, the same reprogrammed, continuously dividing stem cell population may accumulate mutations and progress to **dysplasia** and then carcinoma.
WHY: Metaplasia itself is not malignant and is reversible. What is dangerous is the soil it creates: a proliferating population under continuing stress.

NET RESULT A tissue that has traded its specialised function for durability — protected in the short term, and at increased cancer risk in the long term.

TABLE 3.4 THE METAPLASIAS WORTH KNOWING

Site	Change	Cause	Consequence
Stomach — this session's slide	Gastric → intestinal epithelium with goblet cells	Chronic gastritis, especially <i>Helicobacter pylori</i>	Precursor to gastric adenocarcinoma of intestinal type
Bronchus	Ciliated columnar → stratified squamous	Cigarette smoke	Loss of the mucociliary escalator; precursor to squamous cell carcinoma
Oesophagus (Barrett)	Squamous → columnar with goblet cells	Chronic acid reflux	Precursor to oesophageal adenocarcinoma

Site	Change	Cause	Consequence
Connective tissue	Fibrous tissue → bone or cartilage	Injury, e.g. myositis ossificans after muscle trauma	Mesenchymal metaplasia — the one that is not epithelial

HIGH-YIELD — THE FOUR WORDS IN ONE TABLE

If you remember one thing from Part 3, make it this pairing:

Atrophy = smaller cells · **Hypertrophy** = bigger cells · **Hyperplasia** = more cells · **Metaplasia** = different cells.

Size, size, number, identity. Every exam question on this topic is testing which of those four axes has moved.

PART FOUR

04

This Session's Specimens

Every specimen on the list, on one card: what you see, why it looks that way, what to call it, and the sentence to write. Work each card by covering everything below *What you see* and producing the rest yourself — that is precisely the task you are set at the bench and in the exam.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Describe and diagnose each of the six gross specimens unaided.
- Identify lipofuscin and intestinal metaplasia down the microscope.
- Explain why an obstructed kidney is enlarged and atrophic at the same time.
- Separate the pairs that look alike — atrophy from hypoplasia, hypertrophy from dilatation.

4.1 Gross Specimens — Atrophy

GROSS SPECIMEN NO. 1 · GROSS

Brown atrophy of the heart

WHAT YOU SEE

- The heart is **reduced in size**, and the **transverse diameter is reduced more obviously** than the long axis — so the organ looks narrow and **slender**.
- The colour is **light brown** rather than the normal red-brown.
- The **coronary arteries on the surface are highly meandering and serpentine** — they take a winding, snake-like course.

WHY IT LOOKS LIKE THAT

Two independent consequences of one process.

Because: *The myocytes shrink by autophagy (§3.2), so muscle mass and therefore heart size fall. The lipofuscin left over from that autophagy accumulates in the myocyte cytoplasm and, summed over billions of cells, tints the whole organ brown. Meanwhile the epicardial coronary arteries keep their original length while the heart beneath them contracts — a fixed-length vessel on a shrinking surface has nowhere to go but into curves. The tortuosity is not vascular disease; it is geometry.*

PATHOLOGICAL DIAGNOSIS

Brown atrophy of the heart

WRITE IT LIKE THIS

Organ: Heart. **Description:** The heart is reduced in size, with the transverse diameter reduced disproportionately, giving a slender outline. The colour is light brown throughout. The epicardial coronary arteries follow a markedly tortuous, serpentine course over the surface. No focal lesion, scar or chamber dilatation is seen. **Diagnosis:** Brown atrophy of the heart.

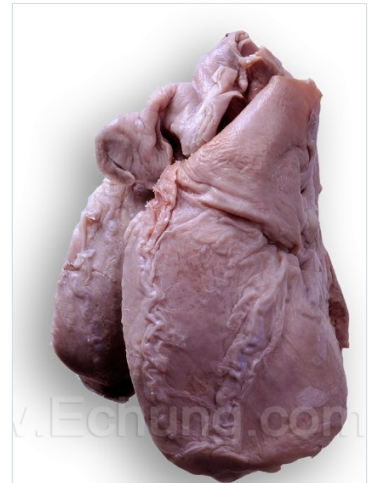


Fig 4.1 Compare the outline with Fig 2.1: this heart is narrow and slender, and the coronary vessels wander instead of running straight.

CLINICAL CORRELATION — WHO THIS HEART BELONGED TO

Brown atrophy is the heart of **old age or chronic wasting** — advanced cancer, tuberculosis, severe malnutrition. The old term is **cardiac cachexia**. It is the opposite end of the spectrum from the specimen in §4.2, and the two are often shown side by side for exactly that reason.

GROSS SPECIMEN NO. 2 · GROSS

Atrophy of the brain

WHAT YOU SEE

- The brain is **reduced in size and weight**.
- The **gyri are narrowed** and the **sulci are correspondingly widened and deepened**.
- The **leptomeninges appear loose** over the surface, and (on a coronal section) the ventricles are dilated to fill the space.

WHY IT LOOKS LIKE THAT

The deck attributes it to **lack of nutrients** — chronic ischaemia and undernutrition of the cortex.

Because: Neurons are **permanent** cells. They cannot be replaced, so loss of neurons and of their dendritic trees is permanent volume loss. The gyri, which are made of cortex, shrink; the sulci between them are spaces, so they can only widen. The skull cannot shrink to match, so the vacated volume is taken up by cerebrospinal fluid — which is why the ventricles dilate. That dilatation is passive and is called **hydrocephalus ex vacuo**: fluid filling a space, not fluid under pressure.



Fig 4.2 Narrowed gyri and correspondingly widened sulci. The brain has pulled away from its own surface.

PATHOLOGICAL DIAGNOSIS

Atrophy of the brain, due to lack of nutrients

WRITE IT LIKE THIS

Organ: Brain. **Description:** The brain is reduced in size. The gyri are narrowed and the intervening sulci are widened and deepened, most evidently over the frontal and temporal lobes. The leptomeninges lie loosely over the shrunken surface. **Diagnosis:** Cerebral atrophy.

GROSS SPECIMEN NO. 3 · GROSS

Renal compression atrophy — obstruction of the ureter

WHAT YOU SEE

- The kidney is **enlarged**, and the **surface is uneven and nodular**.
- On the **cut surface**: the **renal pelvis and calyces are highly dilated**.
- The **renal parenchyma is significantly atrophied and thinned**.
- The **boundary between cortex and medulla is unclear**.

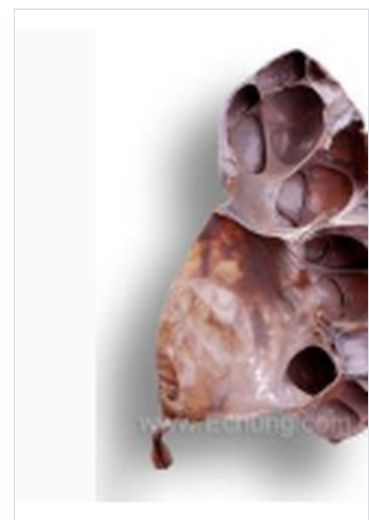


Fig 4.3 The cut surface: dilated pelvis and calyces, with the parenchyma reduced to a thin rim.

WHY IT LOOKS LIKE THAT

An obstruction below the kidney turns the collecting system into a pressurised bag.

Because: Urine is produced continuously but cannot escape past the obstructed ureter. It backs up and the pelvis and calyces dilate — that is why the organ as a whole is larger. The rising pressure then acts inwards on the parenchyma from the medulla outwards, compressing tubules and the small vessels that feed them. Compression is one of the listed causes of atrophy (§3.2), and it acts here by two routes at once: direct mechanical pressure on the tubular cells, and **ischaemia** from compressed vasculature. The pyramids flatten first because they are nearest the dilated calyces, which is why the cortico-medullary boundary blurs. The nodular external surface is the cortex bulging between the underlying dilated calyces.

PATHOLOGICAL DIAGNOSIS

Renal compression atrophy (hydronephrosis) caused by obstruction of the ureter

WRITE IT LIKE THIS

Organ: Kidney. **Description:** The kidney is enlarged with an uneven, nodular external surface. The cut surface shows marked dilatation of the renal pelvis and calyces. The parenchyma is markedly thinned and atrophic, and the cortico-medullary junction is indistinct. **Diagnosis:** Compression atrophy of the kidney (hydronephrosis) secondary to ureteric obstruction.

HIGH-YIELD — AN ORGAN CAN BE BIGGER AND ATROPHIC AT ONCE

This is the single most examined point on this specimen, and it catches people every year. **Atrophy is a statement about the parenchyma, not about the organ's outline.** The hydronephrotic kidney is larger than normal because it contains a bag of urine; the functioning tissue within it has been crushed to a rim. If the question asks whether the kidney is enlarged, the answer is yes; if it asks whether it is atrophic, the answer is also yes; and both are in the same sentence of the specimen description.

GROSS SPECIMEN NO. 4 · GROSS

Leprotic atrophy of the lower extremity

WHAT YOU SEE

- **Shortening and deformation of the toes;** mild thinning of the calf.
- The **skin becomes thinner and smoother** — it has lost its normal texture and appendages.
- The cross-section of the calf shows **thinning of the tibial and fibular cortex.**
- **Muscle mass is reduced or essentially absent, replaced by adipose tissue.**



Fig 4.4 Leg and toes, with a cross-section of the calf displayed alongside. Note the shortened, deformed toes and the thin, smooth skin.

WHY IT LOOKS LIKE THAT

One organism, acting on nerves, producing atrophy of four different tissues.

Because: *Mycobacterium leprae* preferentially invades **peripheral nerves**, because it grows best at the cooler temperatures of the skin and superficial nerves of the limbs. Destroying the nerve removes the trophic and motor supply at a stroke, and each tissue then atrophies for its own reason: **muscle** from denervation and disuse, **bone** from loss of the mechanical loading that maintains it (Wolff's law — bone that is not stressed is resorbed), **skin** from loss of autonomic and sensory innervation. The shortening and deformity of the toes is different in kind: it follows from **sensory loss**, which allows repeated unnoticed trauma, ulceration, infection and resorption of the digits. The fat is not new growth — it simply remains where the muscle has gone.

PATHOLOGICAL DIAGNOSIS

Leprotic atrophy of the lower extremity

WRITE IT LIKE THIS

Organ: Lower limb (leg and foot). **Description:** The toes are shortened and deformed and the calf is mildly thinned. The skin is thin and abnormally smooth. The cut cross-section of the calf shows thinning of the cortex of both tibia and fibula, with marked reduction or loss of skeletal muscle, which has been replaced by adipose tissue. **Diagnosis:** Neurogenic (leprotic) atrophy of the lower extremity.

4.2 Gross Specimens — Hypertrophy

GROSS SPECIMEN NO. 21 · GROSS

Hypertrophy of the heart

WHAT YOU SEE

- The heart is **enlarged** and heavier than normal.
- The **left ventricular wall is thickened** — well above the normal 12–15 mm.
- The **left heart chamber is dilated**.
- The papillary muscles and trabeculae carneae are prominent and thickened.

WHY IT LOOKS LIKE THAT

Sustained overload drove each myocyte to add sarcomeres (§3.3), and the wall thickened as a result.

Because: *Because cardiac muscle is a permanent tissue, the only available response to a raised workload is for existing cells to enlarge — the number of myocytes does not change. The thickened wall is the sum of billions of individually enlarged cells.*

PATHOLOGICAL DIAGNOSIS

Cardiac hypertrophy — left ventricular

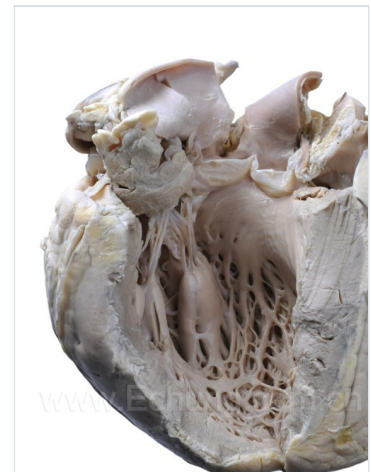


Fig 4.5 The cut left ventricle. Measure the wall against the normal 12–15 mm, and note the trabeculae carneae standing out on the endocardial surface.

WRITE IT LIKE THIS

Organ: Heart. **Description:** The heart is enlarged. On the cut surface the left ventricular wall is markedly thickened, and the papillary muscles and trabeculae carneae are prominent. The left ventricular cavity is dilated. **Diagnosis:** Left ventricular hypertrophy.

EXAM TRAP — THE DECK SAYS “CENTRAL HYPERTROPHY”. READ THIS BEFORE THE EXAM.

The specimen description reads: “the left ventricular wall is thickened, and the left heart chamber is dilated (central hypertrophy)”. Those two features do not normally go together, and the bracket is almost certainly a translation of 向心性肥大, which is standard Chinese for **concentric** hypertrophy — the pressure-overload pattern with a thick wall and a *normal or reduced* cavity. **Eccentric** hypertrophy (离心性肥大) is the volume-overload pattern, with a dilated cavity.

So the safest answer in an exam: **this is left ventricular hypertrophy**, and if you are asked to classify it, say that a thickened wall with a *non-dilated* cavity is **concentric** (pressure overload — hypertension, aortic stenosis), while a thickened wall with a *dilated* cavity is **eccentric** (volume overload — aortic or mitral regurgitation), and state which one the specimen in front of you actually shows. Describing what you see protects you from an ambiguity in the label.

GROSS SPECIMEN NO. 22 · GROSS

Compensatory hypertrophy of the kidney



Fig 4.6 The two kidneys from one patient, shown together. On the left, the enlarged compensating kidney; on the right, the congenitally dysplastic one. The teaching value is entirely in the comparison — neither organ means much alone.

WHAT YOU SEE

- **One kidney is significantly reduced in volume** — congenital dysplasia.
- **The other kidney is increased in volume** — compensatory hypertrophy.
- On the cut surface of the enlarged kidney, the **renal cortex is thickened**.
- The architecture of the enlarged kidney is otherwise **normal** — no dilatation, no scarring.

WHY IT LOOKS LIKE THAT

One kidney is doing the work of two, and has grown to match.

Because: When functioning renal mass is lost, the surviving nephrons receive an increased filtration load. Growth factors and mechanical signals drive the tubular cells to enlarge — and, because tubular epithelium is a **stable** tissue, some of them also divide. The cortex thickens because that is where the tubules are. This is the same principle that lets a living donor give away a kidney: the remaining one enlarges and recovers most of the lost function.

PATHOLOGICAL DIAGNOSIS

Compensatory hypertrophy of the kidney, secondary to contralateral congenital dysplasia

WRITE IT LIKE THIS

Organ: Both kidneys. **Description:** The two kidneys are markedly unequal. One is small and malformed, consistent with congenital dysplasia. The other is uniformly enlarged, with a smooth capsule and no dilatation of the pelvis; its cut surface shows a thickened cortex with preserved cortico-medullary demarcation.

Diagnosis: Compensatory hypertrophy of the kidney, with contralateral congenital dysplasia.

4.3 Microscopic Slides

SLIDE 1 · MICRO

Atrophy of the heart — myocardial brown atrophy

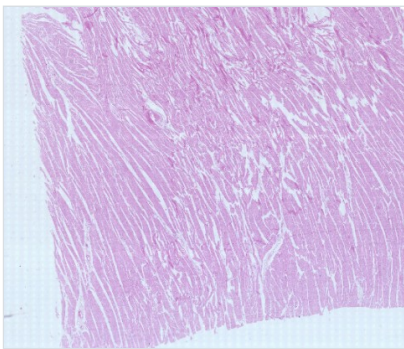


Fig 4.7 4x. Start here. The architecture is preserved — this is still recognisable myocardium — but the fibres are thin and the spaces between them are wide.

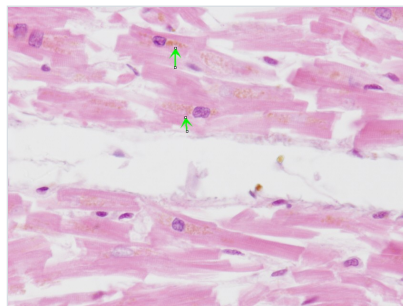


Fig 4.8 40x. The diagnostic finding: **yellow-brown granular pigment** clustered at the poles of the nuclei. Green arrows mark it.

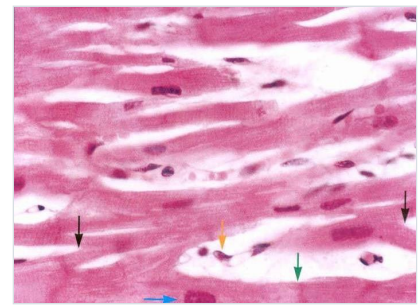


Fig 4.9 40x. Thinned, wavy fibres with widened interstitial spaces between them. Nuclei look crowded because the cytoplasm around them has gone.

WHAT YOU SEE

- **Low power:** myocardial architecture is preserved, but the muscle fibres are **thinner than normal** and the **interstitial spaces between them are widened**.
- **High power:** **yellow-brown granular pigment** in the cytoplasm, concentrated at **both poles of the nucleus**.
- Nuclei appear relatively **crowded and prominent** against the reduced cytoplasm.
- Some **fibre separation and rupture** may be visible.

WHY IT LOOKS LIKE THAT

Each finding is one step of the mechanism in §3.2, made visible.

Because: The *thin fibres* are the autophagy and proteasome degradation of contractile protein. The *widened interstitium* is not new tissue — it is the space vacated as the fibres narrowed, which is why the stroma looks more prominent without having grown. The *pigment* is *lipofuscin*, the indigestible residue of autophagocytosed membrane lipid; it sits at the nuclear poles because that is where the cell's lysosomes and Golgi are concentrated. Because it accumulates over a lifetime and is never cleared, lipofuscin is called the “*wear-and-tear*” *pigment* — its presence dates the process as chronic.

PATHOLOGICAL DIAGNOSIS

Brown atrophy of the myocardium

WRITE IT LIKE THIS

Tissue: Myocardium. **Description:** The overall architecture of the myocardium is preserved. The muscle fibres are thinner than normal and the interstitial spaces between them are correspondingly widened. At high power, fine yellow-brown granular pigment is present in the cytoplasm of many myocytes, concentrated at both poles of the nucleus. No inflammatory infiltrate, necrosis or fibrosis is seen. **Diagnosis:** Brown atrophy of the heart.

BENCH TECHNIQUE — THIS IS THE SLIDE YOU MUST DRAW

Slide 1 is on the homework list as a **drawing**. Draw it at **40×**, not 4×, and include four things: a few thinned muscle fibres with their striations, the widened space between them, a nucleus, and the **granular pigment at the nuclear poles** — drawn as small stippled dots, not as a solid patch. Label the pigment “*yellow-brown granular pigment (lipofuscin)*”. A drawing without the pigment has missed the diagnosis, however neat it is.

SLIDE 2 (SAMPLE SLIDE 84) · MICRO

Gastric mucosa with intestinal metaplasia

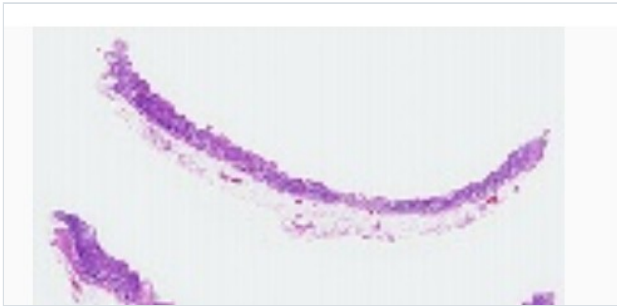


Fig 4.10 4×. A ribbon of gastric mucosa. At this power you establish the organ and see that the surface is intact and the glands are still arranged perpendicular to it.

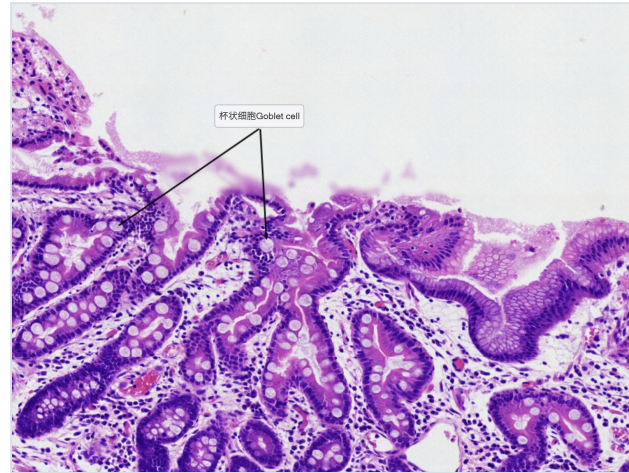


Fig 4.11 40×. The diagnostic cells: **goblet cells** (杯状细胞), labelled, sitting in gastric mucosa where they have no business being.

WHAT YOU SEE

- **Low power:** gastric mucosa, recognised by its surface foveolae and perpendicular glands; the overall architecture is preserved.
- **High power:** scattered among the gastric epithelium are **goblet cells** — tall cells with a clear, mucin-filled cup of cytoplasm pushing the nucleus down to the base.
- The epithelium in those areas looks **intestinal** rather than gastric — a flatter, absorptive-type surface.
- A **chronic inflammatory infiltrate** — lymphocytes and plasma cells — is present in the lamina propria.

WHY IT LOOKS LIKE THAT

The stem cells of the gastric glands have been reprogrammed (§3.5).

Because: Chronic gastritis — very often *Helicobacter pylori* — keeps the mucosa under sustained chemical and inflammatory attack. The gastric stem cells in the gland necks respond by differentiating along an **intestinal** pathway instead, producing goblet cells and absorptive enterocytes that tolerate the environment better. Note what has not happened: no gastric cell has turned into an intestinal cell. The lymphocytes and plasma cells in the lamina propria are the evidence of the chronic irritation that caused it — which is why the slide is properly called gastritis with intestinal metaplasia.

PATHOLOGICAL DIAGNOSIS

Chronic gastritis with intestinal metaplasia

WRITE IT LIKE THIS

Tissue: Gastric mucosa. **Description:** The mucosal architecture is preserved. Within the surface and glandular epithelium there are numerous goblet cells, containing a clear apical mucin vacuole that displaces the nucleus basally, together with epithelium of intestinal absorptive type. The lamina propria contains a chronic inflammatory infiltrate of lymphocytes and plasma cells. No dysplasia or malignancy is identified. **Diagnosis:** Chronic gastritis with intestinal metaplasia.

HIGH-YIELD — THE ONE CELL THAT MAKES THE DIAGNOSIS

A goblet cell in the stomach is intestinal metaplasia. Goblet cells are normal in intestine and normal in bronchus; in gastric mucosa they do not belong, and finding one is the whole diagnosis. Note also the clinical weight of this slide: intestinal metaplasia is a recognised **precursor of intestinal-type gastric adenocarcinoma**, which is why patients who have it are placed under endoscopic surveillance.

4.4 Tell Them Apart — the Look-Alikes

TABLE 4.1 PAIRS THAT ARE CONFUSED, AND THE ONE FEATURE THAT SEPARATES THEM

Pair	They share	The deciding feature
Atrophy vs hypoplasia	A small organ	Did it ever reach normal size? Atrophy: yes, then shrank. Hypoplasia: never did — a developmental defect
Hypertrophy vs hyperplasia	A large organ	Bigger cells or more cells? Decided by the tissue: permanent tissue can only hypertrophy
Hypertrophy vs dilatation	An enlarged heart	Measure the wall. Hypertrophy thickens it; pure dilatation stretches and <i>thins</i> it while the cavity enlarges
Metaplasia vs dysplasia	An abnormal epithelium at a site	Are the cells well differentiated? Metaplasia: yes — orderly, mature, just the wrong type. Dysplasia: no — disordered, atypical nuclei, loss of maturation. Metaplasia is reversible; dysplasia is pre-malignant
Lipofuscin vs haemosiderin	Yellow-brown granular cytoplasmic pigment	Prussian blue stain. Haemosiderin contains iron and stains blue; lipofuscin does not and stays brown. Context helps too: lipofuscin in atrophic heart and liver, haemosiderin after haemorrhage or congestion
Hydronephrosis vs simple renal atrophy	Thinned renal parenchyma	Is the pelvis dilated? Hydronephrosis: yes, and the whole organ is <i>enlarged</i> . Simple atrophy: no, and the organ is small

PART FIVE

05

Homework, Self-Test and Glossary

What you have to hand in, what is coming next week, sixteen questions to check yourself against, and the vocabulary in one place.

5.1 Your Homework — Exactly What to Submit

Take the photographs from the **Digital Human STEM Morphology System** digital slide and organ system, and submit them in a **Word document** on **Xuezaizheda** (学在浙大).

TABLE 5.1 THE SUBMISSION CHECKLIST

#	Item	Task and where it is covered here
1	Normal organ GROSS — heart, lung <i>or</i> kidney (choose one)	Describe. Use the six words. §2.1–2.3
2	Atrophy of the heart GROSS	Describe. §4.1, Specimen 1
3	Atrophy of the kidney caused by obstruction GROSS	Describe. §4.1, Specimen 3
4	Atrophy of the heart MICRO — Slide 1	Draw at 40×, with the pigment labelled. §4.3, Slide 1

BENCH TECHNIQUE — A REPORT THAT MARKS WELL

- Use the department's four headings for **every** item: specimen number and type · organ or tissue · description (evidence of diagnosis) · pathological diagnosis.
- Keep the description in **observable language** — no diagnostic words above the diagnosis line (§1.4).
- Include at least one **negative** finding per item. It shows you looked for the alternatives.
- For item 1, describe the organ you have actually photographed — do not copy a textbook description of a different heart.

5.2 Next Week — What Is Coming

The stated purpose of next week is the same as this week's — *master the concepts of hypertrophy, hyperplasia and metaplasia* — plus one new pair: **list the morphological appearances of oedema and hydropic change, and their related organs**. In other words, the course now moves from **adaptation** to **reversible cell injury**.

TABLE 5.2 NEXT WEEK'S SPECIMEN LIST

Type	Specimen	What to revise beforehand
GROSS	No. 5 — Cloudy swelling of the liver	New. Hydropic change: the organ is enlarged, pale and firm with a dull cut surface
GROSS	No. 21 — Hypertrophy of the heart	Already covered — §4.2
GROSS	No. 22 — Hypertrophy of the kidney	Already covered — §4.2
MICRO	Slide 9 — Cloudy swelling of the kidney	New. Swollen tubular cells with granular cytoplasm and narrowed lumina
MICRO	Slide (84) — Gastric mucosa with intestinal metaplasia	Already covered — §4.3, Slide 2

HIGH-YIELD — THREE OF FIVE ARE ALREADY IN THIS GUIDE

Only **cloudy swelling** is genuinely new next week. If you learn Part 4 of this guide properly now, three of next week's five specimens are already done — and the concepts Part 3 gives you are the same ones next week's stated purpose asks for.

5.3 Self-Test — 16 Questions with Answers

Short-answer, in the format this laboratory is actually examined in. Cover the answer, produce yours, then compare. The answers are deliberately written as sentences you could hand in.

Q1 In what order do you describe a gross specimen, and why that order?

Organ → whole organ → lesion. The order moves from certain to uncertain: you can identify the organ before you can characterise its disease, and a description that begins at the lesion has no frame of reference to measure the lesion against.

Q2 Name the six features you record for the whole organ.

Size · weight · shape · colour · surface · cut surface.

Q3 Why must you scan a slide at 4× before going to 40×?

Because **the diagnosis is made on architecture**, which is only visible at low power. At 40× the field is too small to show the pattern; if a process is diffuse, everything looks uniform and the abnormality becomes invisible.

Q4 Give the normal weight of the heart and of one kidney.

Heart: about **280 g** in a man, **240 g** in a woman. Kidney: **125–150 g** each.

Q5 Name the three layers of the heart wall from inside out, and give one histological feature that identifies each.

Endocardium — endothelial lining with a subendothelial layer containing Purkinje fibres (large, pale, glycogen-rich cells). **Myocardium** — striated, *branching* muscle with central nuclei. **Epicardium** — mesothelium over connective tissue containing **adipose tissue** and vessels.

Q6 Define atrophy.

A **decrease in the size** of a cell, tissue, organ or multiple organs, associated with pathological conditions such as ischaemia, malnutrition or hormonal change. Note that it is primarily a reduction in cell *size*; cell number may also fall, but the atrophic cell is alive.

Q7 Which two cellular machines remove bulk during atrophy, and which one produces lipofuscin?

The **ubiquitin–proteasome pathway** (cytosolic proteins) and **autophagy** (organelles). **Autophagy** produces lipofuscin — it is the undigestible residue of autophagocytosed membrane lipid.

Q8 Why are the coronary arteries tortuous in brown atrophy of the heart?

Because the arteries keep their original length while the heart beneath them shrinks. A fixed-length vessel on a contracting surface must take up the slack in curves. It is **geometry, not vascular disease**.

Q9 A kidney is larger than normal yet the diagnosis is atrophy. Explain.

Ureteric obstruction dilates the pelvis and calyces with retained urine, so the **organ** enlarges. The rising pressure compresses the parenchyma — mechanically and by causing ischaemia — so the **functioning tissue** atrophies to a thin rim. Atrophy describes the parenchyma; enlargement describes the outline. Both are true.

Q10 Distinguish atrophy, hypoplasia and aplasia.

Atrophy — the organ reached normal size, then shrank. **Hypoplasia** — it never reached normal size (developmental). **Aplasia** — it never formed. In the compensatory renal specimen, the small kidney is **hypoplastic**, not atrophic.

Q11 Why can the heart hypertrophy but never hyperplase?

Cardiac muscle is a **permanent** tissue: its myocytes left the cell cycle before birth and cannot re-enter it. Enlargement of existing cells is the only growth mechanism available, so increased workload can only thicken cells, never multiply them.

Q12 Concentric or eccentric — which follows pressure overload, and what does the cavity look like?

Concentric. Sarcomeres are added in parallel, the wall thickens, and the cavity is **normal or reduced**. Eccentric hypertrophy follows **volume** overload: sarcomeres in series, a **dilated** cavity. In one line: pressure thickens, volume dilates.

Q13 Define metaplasia. Which cells actually change?

A **reversible** change in which one differentiated cell type is replaced by another better suited to the environment. **No mature cell transforms.** The **tissue stem cells are reprogrammed** to differentiate along a new pathway.

Q14 A goblet cell is seen in gastric mucosa. What is the diagnosis, the likely cause, and the clinical significance?

Intestinal metaplasia, in the setting of chronic gastritis — very often *Helicobacter pylori*. It is a recognised **precursor of intestinal-type gastric adenocarcinoma**, so affected patients are kept under endoscopic surveillance.

Q15 How do you distinguish lipofuscin from haemosiderin?

A **Prussian blue** stain. Haemosiderin contains iron and turns blue; lipofuscin contains none and remains brown. Context supports it: lipofuscin in the atrophic heart, liver and neurons; haemosiderin after haemorrhage or chronic congestion.

Q16 Why is metaplasia both protective and dangerous?

Protective — the replacement epithelium is tougher and lets the tissue survive a hostile environment. **Dangerous** — the original specialised function is lost, and if the irritant persists, the continuously proliferating reprogrammed stem cells may accumulate mutations and progress through dysplasia to carcinoma.

5.4 Glossary of Key Terms

Anthraxis

Black pigmentation of the lung from inhaled carbon. Normal in any adult city dweller and not, by itself, a disease.

Aplasia

Complete failure of an organ to develop. Contrast hypoplasia (developed, but incompletely) and atrophy (developed fully, then shrank).

Atrophy

Decrease in the size of a cell, tissue or organ, by reduced synthesis and increased degradation of cell contents. The cell survives at a lower metabolic level; it is an adaptation, not a form of death.

Autophagy

The cell enclosing its own organelles in vacuoles and digesting them with lysosomal enzymes. The main route by which an atrophic cell sheds bulk — and the source of lipofuscin.

Cardiac cachexia

Older term for brown atrophy of the heart in chronic wasting disease.

Cloudy swelling

Hydropic change: an early, **reversible** injury in which cells swell with water because ATP-dependent ion pumps

fail. The organ is enlarged, pale and firm with a dull cut surface. *Next week's topic.*

Compensatory hypertrophy

Enlargement of a surviving organ or tissue after loss of its partner or of part of itself — as in the remaining kidney after unilateral loss.

Concentric hypertrophy

Thickened wall with a **normal or reduced** cavity, from **pressure** overload. Sarcomeres are added in parallel.

Cut surface

The face exposed when a specimen is sectioned. Described separately from the external surface, because most of the informative pathology of a solid organ is internal.

Dysplasia

Disordered growth with **atypical nuclei and loss of orderly maturation**. Unlike metaplasia it is **pre-malignant**. The two are distinguished by whether the cells are well differentiated.

Eccentric hypertrophy

Thickened wall with a **dilated** cavity, from **volume** overload. Sarcomeres are added in series.

Endocardium

The innermost layer of the heart: endothelium over a subendothelial layer of smooth muscle containing **Purkinje fibres** of the conducting system.

Epicardium

The outermost layer: **mesothelium** (simple squamous epithelium) over thin connective tissue containing vessels, nerves and **adipose tissue**. The fat is the quickest way to identify it.

Goblet cell

A mucin-secreting cell with a clear apical cup that pushes the nucleus basally. Normal in intestine and bronchus; **in gastric mucosa it is diagnostic of intestinal metaplasia**.

Gross specimen

An organ or tissue examined with the naked eye — “macroscopic”. Described by size, weight, shape, colour, surface and cut surface.

Haemosiderin

Iron-containing golden-brown pigment derived from breakdown of haemoglobin. Distinguished from lipofuscin by staining **blue with Prussian blue**.

Hydrocephalus ex vacuo

Dilatation of the ventricles that simply fills space vacated by cerebral atrophy. Passive, not caused by raised pressure.

Hydronephrosis

Dilatation of the renal pelvis and calyces from obstruction to urine outflow, with secondary **compression atrophy** of the parenchyma. The organ is enlarged while its functioning tissue is reduced.

Hyperplasia

Increase in the **number** of cells by proliferation. Possible only in labile and stable tissues; impossible in permanent tissue.

Hypertrophy

Increase in the **size** of cells with no increase in their number, enlarging the organ. The only growth option available to permanent tissue.

Hypoplasia

Failure of an organ to reach normal size during development. Congenital — distinguishing it from atrophy, which is acquired.

Interalveolar septum

The wall between two alveoli, containing the alveolar capillaries. Normally hair-thin; any thickening is abnormal.

Lipofuscin

Yellow-brown, granular, **iron-negative** pigment: the undigestible residue of autophagocytosed membrane lipid. Accumulates at the nuclear poles over a lifetime, hence the “wear-and-tear pigment”. Its presence marks a process as chronic.

Medullary ray

The straight bundle of collecting tubules running from the medulla into the renal cortex. A useful landmark for orienting a renal slide.

Metaplasia

Reversible replacement of one differentiated cell type by another better able to withstand the local environment, by **reprogramming of tissue stem cells**. Protective in the short term; a risk factor for malignancy if the stimulus persists.

Permanent tissue

Tissue whose cells have left the cell cycle and cannot re-enter it: **cardiac muscle, skeletal muscle, neurons**. Can hypertrophy; cannot hyperplase or regenerate.

Purkinje fibre

Modified cardiac muscle cell of the conducting system, found in the subendocardial layer. **Large, pale and glycogen-rich** — bigger and emptier-looking than a working myocyte.

Stable tissue

Tissue normally quiescent but able to re-enter the cell cycle when required: liver, renal tubular epithelium, fibroblasts, smooth muscle. Can both hypertrophy and hyperplase.

Ubiquitin–proteasome pathway

The system that tags unneeded cytosolic proteins with ubiquitin and destroys them in the proteasome. The main route by which disused muscle loses its contractile protein.

Wolff’s law

Bone remodels in response to the mechanical load placed on it. Unloaded bone is resorbed — which is why a denervated or immobilised limb shows cortical thinning.